William Tucker

Software Engineer · Student Data Pipelines on Oracle and PostgreSQL · 13+ Years at BC Universities · Kelowna, BC

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PROFESSIONAL SUMMARY

Software engineer with **over 13 years** on the systems that hold student data at BC post-secondary institutions. At Vancouver Island University I maintain the data warehouse pipeline behind the university's statutory submission to the province, built VIU's integration with StudentAid BC's SIMS platform, and designed the team's DevOps workflow and onboarding and reference handbook. Deep SQL (Oracle PL/SQL, PostgreSQL); Python for scanners, ETL, and tooling.

I was programming for a decade before AI tooling existed, so I oversee every task it performs with careful scrutiny, using custom tooling I built; the design, review, tests, and production accountability stay with me. Everything I build ships with documentation and training from a system I designed.

Production data pipelines | Advanced SQL (Oracle, PostgreSQL) | Sensitive and regulated student data | Access control and privacy | Data quality and reconciliation | Documentation and runbooks | AI-native development and review

TECHNICAL SKILLS

Data & SQL	Oracle (PL/SQL, schema design, views and grants, Data Pump, tuning), PostgreSQL (Supabase, Row-Level Security, pgvector), versioned migrations with verify and rollback scripts
Pipelines	Daily file-based extract-and-load (fixed-width government feeds, idempotent loads, automatic backfill), bulk exports, staged-then-published loads, SFTP/SSH, REST
Scheduling	Oracle DBMS_SCHEDULER, Windows Task Scheduler, GitLab and GitHub Actions schedules, n8n.
Data Quality	Golden-file tests, old-vs-new output diffs before go-live, source row-count checks, missed-run alerts, short-load refusal, read/write lineage mapping
Privacy & Access	Least-privilege views, Row-Level Security, segregation of duties, masked identifiers, PII-free audit logging, SSO/SAML/OAuth
Languages & DevOps	SQL, PL/SQL, Python, C#, TypeScript, Go, PowerShell · GitLab CI/CD, GitHub Actions, IIS, Pester, Docker Compose
AI Tooling	Claude Code daily; my own documentation system, workflow plugin, and session coordinator (below); custom MCP servers; RAG with pgvector

EXPERIENCE

Programmer/Analyst, Vancouver Island University

Jul 2016 – Present · Nanaimo, BC (remote from Kelowna)

Senior engineer on an Enterprise Systems team of 18. I own the team's deployment workflow, maintain the provincial reporting pipeline, and build the systems that move student data, each scoped with a business analyst and tested by the departments that own the data.

Central Data Warehouse: provincial student-data submission

- **Maintain the pipeline** that sends VIU's student-level data (about 2.2 million course registrations and 260,000 student records per cycle) to the Ministry twice a year: a nightly 34-table Oracle 19c rebuild, a PL/SQL extract into 14 Ministry tables, and a Data Pump export. I own the PL/SQL, scheduler jobs, storage, and credentials; first worked on it in 2017.

- **Designed and built the CDW Submission Tool** (.NET 8, production, 2025–present), automating much of the DBA work behind each submission: it runs the Data Pump export of all 14 tables, blocks submission until six pre-flight health checks pass, and prevents duplicate submissions per period. Adapted it when the Ministry moved delivery to SharePoint in 2026.
- **Support the department through each cycle.** I troubleshoot and fix the pipeline, keep storage and credentials healthy, coordinate with the contractor who refreshes the reporting clone, and wrote the runbooks and troubleshooting guides staff work from.

StudentAid BC enrolment confirmation (production, 2026)

- **VIU's SIMS platform integration tool.** Designed and built in two months as sole engineer. When StudentAid BC moved every BC institution onto SIMS in June 2026, it replaced an undocumented legacy Oracle package written by staff no longer at VIU. The system is how VIU confirms each student's enrolment, which is the step that releases their loan and grant funding.
- **End to end, every day, unattended.** Imports the province's nightly 121-field applicant feed and confirmation queue into Oracle, never loading a file twice and catching up missed days on its own. Financial Aid staff work a review screen that suggests each decision with a plain-language reason and masks SINs. The outbound file is held byte-exact to the province's format by golden-file tests. Peak day: 3,425 records in, 2,958 confirmations out.
- **Built to report its own failures.** Daily success emails, health endpoints, an off-server watchdog, and over 800 CI-gated tests. Worked directly with the province's analyst on format fixes, and reconciled every post-launch issue affecting funding with a documented no-loss finding.

Team DevOps workflow and handbook

- **Moved about 90 applications** off in-place editing on a network share onto git and GitLab CI/CD (Oct 2025): a branching strategy behind a manual production gate, shared pipeline templates, automated tests, rollback scripts, and Teams deploy notifications, adopted by a team of about 6–8 developers.
- **Wrote the Enterprise Systems Handbook.** The team's onboarding and reference guide to how we work: branching strategy, deploy and rollback strategy, CI/CD, environment and system configuration, server setup, platform architecture, Oracle schemas, and SSO. The team of 18 onboards from it and I keep it current.

Modernizations

- **Atlas** (dev, 2026). Modernization planning across about 150 applications, with a Python scanner that maps every read and write to an Oracle schema down to file and line (7,152 references for one schema).
- **BIS, Building Inventory System** (production, 2026). Rebuilt from Classic ASP to .NET 8 Blazor on Oracle with QuestPDF reports. Its modernization guide is now VIU's playbook for converting the rest of the legacy portfolio.
- **LaunchPad / Portal** (dev, 2026). .NET 8 Blazor replacement for the university's SSO landing page: direct SAML login, impersonation with privilege checks, the app catalog, and a live team activity feed. Fixed a privilege-escalation bug where identity came from a client-writable cookie.

CTO and Co-founder, AscendAI Technologies

Apr 2026 – Present · Kelowna, BC (alongside VIU)

- **AI land development platform for BC** (pilot). Feasibility assessments, multi-parcel assembly analysis, and bylaw and feasibility chatbots, with every figure cited to the bylaw it came from.
- **Bylaw ingestion and parsing engine.** Turns municipal bylaws into structured, citable data for our RAG system, verified before anything is published.
- **User-generated reports** built from the same verified source, secured with Row-Level Security and a tamper-evident audit log.

Tooling I built for AI-assisted development

2026 · used daily at VIU and AscendAI

- **wts-ai-docs.** Documentation written for AI agents, with a CI gate that blocks code changes without doc updates. In a controlled test, agents answered 45% faster with 68% fewer tool calls and zero wrong answers.

- **Cadence.** A Claude Code plugin in Go that enforces my development workflow so an agent can't skip steps.
- **Huddle.** Coordinates several AI coding sessions working in parallel on one codebase.

Software Analyst, Thompson Rivers University

May 2012 – Jul 2016 · Kamloops, BC

- **Employee Survey system** (Grails, Oracle) with reports and ETL across Finance, HR, Payroll, and Student departments.
- **Student ID Card system** in Java, used by thousands of students and integrated with BC Transit, the City of Kamloops, and the Kamloops Canada Games Pool.
- **Institutional Data Analyst** (co-op), where I began. Oracle reporting in SQL and PL/SQL.

OFF THE KEYBOARD *(I also get along with humans)*

- Work with teams from different disciplines to gather requirements and meet acceptance criteria; comfortable saying "not yet" and explaining why.
- Ran SQL and reporting training at TRU so departments could self-serve.
- Presented at the BCNET 2015 conference to an audience of over 100 on building institutional applications with Groovy Grails on the Banner ERP.
- Welcomed new developers at TRU with campus tours and introductions to the people and departments they'd be working with.

EDUCATION

Computer Science Diploma, Thompson Rivers University — 2010 – 2013

Journeyman Marine Technician, BCIT · *prior trade career* — 2003 – 2007

References available on request, including current employer.